My Stack **July 2026** **Peptides and Supplements**

Function	***Name+B3:E30***	***Dose***	***Frequency***	***Pattern***
Metabolic	**Retatrutide**	SC 2.4 mg	Sun, Wed	Continuous
	Metformin	PO 500 mg	Daily	Continuous
GH axis	**Ipamorelin + CJC-1295 no-DAC**	SC 200 mcg + 200 mcg	Sun–Thu PM (fasted)	12 weeks on / 6 weeks off
	Tesamorelin + Ipamorelin	SC 2 mg + 200 mcg	Mon-Fri AM (fasting)	
Tissue repair	**GHK-Cu**	SC 1 - 2 mg	Daily	Alternating vials
	GLOW	SC 1.2 - 2.4 mg	Daily	
	ARA-290	SC 0.5 mg	Daily	Continuous
	Zinc Picolinate	PO 50 mg	3× weekly	Continuous
Anti-inflammatory	**KPV**	SC 750 mcg	Daily	Continuous
	Omega-3	PO DHA 900 mg; DPA 140 mg	Daily, w food	Continuous
Immune modulation	**Thymosin-α1**	SC 0.8 mg	M, F	Continuous
	Thymulin	SC 0.5 mg	Sn, Tu, Tr	Continuous
Longevity	**FOX-04-SRI**	SC 25 mg	Q48H ×3	Every 3 years
	Epitalon + Crystagen + Pinealon	SC 5 mg; 0.5 mg; 0.5 mg	Daily for 10 days	Annually, other peptides paused
	Spermidine 3HCL	PO 10 mg	Daily, evenings	Continuous
	Tadalafil	PO 5 mg	Daily	Continuous
Prostrate support	**Prostamax (KEDP)**	SC 0.5 mg	Daily for 10 days	Annually
	Beta Sitosterol	PO 250 mg	Daily, w food	Continuous
Neurotrophic	**Semax**	IN 600 – 900 mcg	Daily	Occasionally
Sleep support	**Epitalon + DSIP**	IN 200 + 200 mcg	Daily 1 hour before bedtime	Continuous
	L-Glycine	PO 3000 mg		Continuous
Cellular energy & Brain support	**NMN**	PO 1000 mg	Daily	Continuous
	Acetyl-L-Carnitine (ALCAR)	PO 1000 mg	Daily	Continuous
	CoQ10	PO 200 mg	Daily, w food	Continuous
	Creatine HCl	PO 1500 mg	Daily	Continuous
	PQQ	PO 20 mg	Daily	Continuous
	Apigenin	PO 200 mg	Daily, w food	Continuous
	Beta-Alanine	PO 3000 mg	Daily	Continuous

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment.

Updated: 9-Aug-26

Before You Stack

Peptides and supplements are multipliers of fundamentals: good sleep, good nutrition, and exercise. If your sleep is poor, your nutrition is fast food, your stress is chaotic, or you're not lifting, nothing in a jar, bottle, or vial is going to help much. Get your fundamental right, then consider your goals and how supplements, peptides, etc. might help.

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

This document was updated 9-Aug-26. The most recent version of this document is available at https://pdfs.researcher6076.com/pdfs/my-stack.pdf

Lifestyle Stack

Nutrition: I am 100% vegan, focused on whole foods, and limiting processed food.

Intermittent Fasting: I fast 6 days a week from 7 PM until 11 AM.

Sleep: I am focusing on sleep improvements. My bedroom is dark; the bed is temperature controlled for a cool sleep. My wake time is the same each day. My average sleep duration is 8 hours at least 5 days/week, but, my sleep still needs improvement.

Exercise: My resistance training schedule is two – three times each week, consistently. The training is a complete upper and lower body routine with each movement focusing on maximum intensity with 8 – 20 reps. When 20 reps are consistently achieved, I increase the resistance.

Stack Goals

My overarching goal is **health support**. The individual goals supporting this are **metabolic stability**, **body recomposition**, **recovery**, **aging support & healthspan**. As this stack pursues these individual goals it **never intentionally wavers from the healthiest choice**.

Functional Benefits

- **Improved glycemic control and appetite suppression** via GLP-1/GIP signaling.

- **Increased resting energy expenditure and visceral fat reduction** through glucagon receptor activity and GH-mediated lipolysis.
- **Enhanced recovery, sleep quality, and anabolic signaling** from pulsatile GH stimulation.
- **Accelerated tissue repair and collagen support** for skin, tendon, ligament, and soft tissue.
- **Microvascular protection** improving perfusion during injury.
- **Reduced gut and systemic inflammation and improved epithelial repair** supporting GI comfort and mucosal healing.
- **Steadier, more youthful immune tone** with improved antiviral, Th1/NK activation, T-cell regulation, and anti-inflammatory balancing
- **Cognitive enhancement and neuroplasticity** with improved focus and stress resilience.
- **Mitochondrial support, NAD+ restoration, and enhanced mitochondrial biogenesis** through NMN, ALCAR, CoQ10, creatine, Beta-Alanine, and PQQ-mediated redox signaling
- **Mitochondrial support and NAD+ restoration** to support cellular energy and recovery.
- **Improved exercise performance and buffering capacity** for high-intensity efforts.
- **Potential longevity and circadian benefits** including **speculative** telomeric and pineal effects (from Epitalon and Crystagen supported by limited preclinical and clinical data.)
- **Targeted senescent-cell clearance** promoting selective removal of damaged, non-dividing cells, lowering SASP-driven inflammation, freeing tissues from pro-inflammatory, repair-inhibiting signals
- **Autophagy and mitophagy enhancement** stimulating cellular cleanup and mitochondrial quality control.
- Support **prostate tissue repair, reduce local inflammation, and improve microcirculation** (supported by limited preclinical and clinical data.)

Metabolic Benefits

The stack is designed to produce **robust metabolic improvement** by combining appetite suppression, incretin-mediated insulin support, and increased energy expenditure. **Retatrutide** provides strong **GLP-1/GIP**-driven reductions in post-prandial glucose and appetite. Its **glucagon-receptor** activity that increases **resting energy expenditure and fat oxidation**, particularly in visceral and hepatic depots. **Metformin** supports hepatic insulin sensitivity and reduces gluconeogenesis, while **NMN, ALCAR, creatine, and Beta-Alanine** support cellular energy and exercise capacity, indirectly improving metabolic flexibility.

PQQ enhances mitochondrial biogenesis and redox cycling, improving metabolic flexibility and supporting higher resting energy expenditure. Together these agents produce **reduced caloric intake, improved glycemic control, and greater caloric burn**, favoring body composition improvement.

GH Benefits

With the GH axis regimen (Ipamorelin + CJC-1295 NODAC before sleep and Tesamorelin + Ipamorelin upon waking, both 12-weeks ON, 6-weeks-OFF), I should expect **a measurable increase in pulsatile GH secretion and a rise in circulating IGF-1** compared with baseline. In practical terms, over time this stack is likely to shift average GH/IGF-1 markers toward values more typical of **middle-aged adults** rather than untreated older adults, depending on baseline, dose response, and individual variability. Exact numeric increases vary widely by individual; monitoring IGF-1 is the reliable way to quantify the effect.

Regenerative Benefits

This stack combines peptides that act at **complementary points in the repair cascade** to **speed matrix remodeling, reduce local inflammation, and restore tissue architecture**. **GHK-Cu upregulates collagen and proteoglycan synthesis** and modulates MMP/TIMP balance to rebuild extracellular matrix; **TB-500 enhances actin dynamics, cell migration, and angiogenesis** to mobilize reparative cells and revascularize injured tissue; **BPC-157 stimulates local growth-factor signaling and epithelial restitution** to stabilize microcirculation and accelerate tendon, ligament, and gut mucosal healing; and **KPV reduces NF-κB driven cytokine release and supports epithelial barrier repair**, lowering the inflammatory milieu that impairs regeneration.

ARA 290 complements GHK Cu/TB500/BPC 157 by **protecting microvasculature and reducing local inflammation** that otherwise impairs matrix remodeling.

The GH-axis agents (**Ipamorelin + CJC-1295 no-DAC and Tesamorelin**) provide systemic support by increasing pulsatile GH/IGF-1 signaling, which **promotes protein synthesis, satellite-cell activation, and anabolic repair processes** to amplify the local actions of the regenerative peptides.

Prostamax (KEDP) is speculative, with little human evidence for efficacy. It may support **local prostate tissue regulation and microcirculation**, reduce inflammation, and promote tissue homeostasis.

Immune Benefits

This stack enhances an aging immune system by combining **Tα1-driven activation of dendritic cells, NK cells, and interferon-regulated Th1 pathways. Tα1's long-tail signaling** maintains elevated antiviral gene expression, improved antigen presentation, and stronger cytotoxic readiness.

Thymulin-mediated tuning of thymocyte maturation, T-cell activation thresholds, and suppression of pro-inflammatory cytokines like TNF-α, IL-1β, and IL-6, creating a more **coordinated, low-noise immune environment**. Its **shorter regulatory window** repeatedly recalibrates T-cell sensitivity, enhances thymic-hormone signaling, and stabilizes cytokine ratios.

Because Thymulin **improves the *quality and balance* of the T-cell pool** and Tα1 **improves the *effectiveness and readiness* of those cells once deployed**, their mechanisms produce a more coherent, youth-like immune response with less inflammatory noise and more efficient antiviral and cytotoxic function. The result is a **youth-like immune rhythm** characterized by cleaner innate–adaptive crosstalk, reduced inflammatory drift, more efficient T-cell recruitment, and a more predictable, resilient response to immune stressors.

Sleep Benefits

Epitalon, DSIP and L-Glycine, combined with CGC-1295 NODAC + Ipamorelin, together leverage **distinct biological layers of sleep regulation and regeneration:** In combination, these agents support a biologically coherent sleep environment characterized by **stronger circadian entrainment**, **reduced nighttime sympathetic tone**, **enhanced slow-wave depth**, and **more efficient overnight regeneration**.

Epitalon supports sleep by modulating **pineal rhythmicity and melatonin-linked circadian signaling**, helping stabilize the suprachiasmatic nucleus and reinforce the timing cues that govern sleep–wake transitions. **DSIP** contributes through its influence on **stress-responsive neuroendocrine pathways**, including attenuation of **CRH-driven arousal** and modulation of **GABAergic and serotonergic tone**, which together promote smoother entry into slow-wave sleep and more consolidated sleep cycles. **L-Glycine** complements these effects by acting as an **inhibitory neurotransmitter at NMDA/glycine co-receptors** and by triggering **peripheral vasodilation that lowers core body temperature**, a key physiologic signal for sleep initiation. Evening **Ipamorelin + CJC-1295 no-DAC** amplifies the body's natural **nocturnal GH pulse**, supporting **slow-wave sleep**

architecture, **tissue repair**, and **overnight anabolic recovery**. **Tadalafil**, **Prostamax**, and **Beta Sitosterol** reduce nocturia improving sleep quality.

Healthspan Benefits

A *conserved pathway of aging* is a biological process that shows up across many species, changes predictably with age, and when experimentally modified can speed up or slow down aging. These pathways represent the deep, evolutionarily preserved mechanisms that drive cellular decline, tissue dysfunction, and loss of resilience over time.

If a therapy targets a **conserved pathway of aging** and ***if interventions on that pathway reliably slow aging in model organisms***, then modifying that same pathway in humans is **plausibly healthspan-promoting**, even if direct human longevity data are not yet available.

The following conserved pathways are the **dominant, consensus model** used across ageing biology, geroscience, biotech, and translational longevity research.

Mitochondrial dysfunction — age-related decline in mitochondrial energy production, increased ROS, and impaired metabolic flexibility.

Deregulated nutrient sensing (mTOR/AMPK/insulin/IGF) — disrupted balance between growth and repair signaling that skews metabolism toward aging rather than maintenance.

Cellular senescence and SASP — accumulation of damaged, non-dividing cells that secrete inflammatory factors and impair tissue function.

Genomic instability and DNA repair — increasing DNA damage and reduced repair capacity that undermine cellular integrity and accelerate aging.

Epigenetic drift — progressive, disorganized changes in DNA methylation and chromatin structure that alter gene expression and degrade cellular identity.

Stem cell exhaustion and regenerative niche — depletion or dysfunction of tissue-specific stem cells, reducing the body's ability to repair and regenerate.

Microvascular decline and endothelial dysfunction — impaired blood flow, nitric-oxide signaling, and capillary density that limit oxygen and nutrient delivery.

Dysbiosis and barrier dysfunction — age-related shifts in the gut microbiome and weakening of epithelial barriers that increase inflammation and metabolic stress.

The following table evaluates the impact of this stack on these conserved pathways of ageing.

Conserved pathway	What you are doing to address it	Coverage
Chronic inflammation (inflammaging)	KPV; ARA-290; omega-3 (DHA/DPA); metformin; Thymosin-α1; Thymulin	**Excellent**
Proteostasis and autophagy	16-hour TRF; spermidine; apigenin; metformin; NMN; resistance training	**Excellent**
Mitochondrial dysfunction	NMN; ALCAR; CoQ10; creatine HCl; Beta-Alanine; PQQ; Retatrutide; exercise	**Excellent**
Deregulated nutrient sensing (mTOR/AMPK/insulin/IGF)	Metformin; Retatrutide; TRF; timed GH-axis cycling	**Good**
Cellular senescence and SASP	Thymosin-α1; Thymulin; apigenin; immune surveillance; FOXO4-DRI senolytic pulses (Q48H ×3 every 2 years)	**Excellent**
Genomic instability and DNA repair	NMN (NAD+ support); good sleep/circadian hygiene; annual Epitalon + Crystagen + Pinealon window	**Good**
Epigenetic drift	Annual Epitalon; resistance training 2×/week; sleep/circadian practices; NMN	**Moderate**
Stem cell exhaustion and regenerative niche	GH-axis peptides (Ipamorelin/CJC-1295, Tesamorelin); GHK-Cu; TB500; BPC-157; exercise	**Moderate**
Microvascular decline and endothelial dysfunction	ARA-290; tadalafil 5 mg daily; omega-3; Retatrutide metabolic benefits	**Excellent**
Dysbiosis and barrier dysfunction	BPC-157; KPV; metformin (indirect); whole-food vegan diet	**Moderate**

Stack Design - Synergies

- **PM Ipamorelin + CJC-1295 no-DAC; AM Tesamorelin + Ipamorelin:** The evening Ipamorelin/CJC-1295 pairing reinforces the natural nocturnal GH surge and **supports recovery-driven signaling during early sleep**. The morning Tesamorelin + Ipamorelin combination produces a distinct, GHRH-dominant GH pulse that **favors daytime lipolysis and visceral-fat mobilization**. Because Tesamorelin (GHRH-R agonist) and Ipamorelin (GHSR-1a agonist) act through independent receptors, **the morning stack generates a synergistic GH response without overlapping with the nighttime CJC-1295-primed pulse**. Together, the two windows create complementary GH stimulation: recovery-oriented signaling at night and metabolic-targeted signaling during the day.

- **Retatrutide + Tesamorelin:** Retatrutide's thermogenic and hepatic fat effects plus , **reduction**.
- **GHK-Cu + TB-500 + BPC-157 + KPV:** regenerative peptides combine **collagen synthesis, actin remodeling, angiogenesis, epithelial repair, and anti-inflammatory signaling**, producing faster soft-tissue and mucosal healing than any single agent alone.
- **GHK-Cu + Zinc Picolinate:** GHK-Cu supplies bioactive copper for repair enzymes while zinc picolinate stabilizes the copper–zinc metalloprotein network, so together they keep metal homeostasis balanced and amplify collagen-driven tissue regeneration.
- **KPV, BPC-157, metformin, and GLP-1 agonists** all effect **gut motility and microbiome shifts for potential additive GI side effects.**
- **Epitalon, DSIP, and L-Glycine:** This combination supports sleep through **complementary neurobiological pathways** for **pineal/circadian signaling**, **sleep-architecture and stress-response pathways**, and **thermoregulatory cooling and inhibitory neurotransmission**.
- **NMN + ALCAR + CoQ10 + Creatine + Beta-Alanine + Apigenin + PQQ:** these mitochondrial and energy supplements synergize to **support NAD+ pools, electron-transport chain efficiency, reduce oxidative stress, fatty-acid transport, rapid ATP regeneration, PQQ-driven mitochondrial biogenesis and intracellular buffering**, improving endurance, recovery, and cellular energy capacity..
- **Semax + NMN/ALCAR:** Semax's neurotrophic effects pair with mitochondrial support to enhance **cognitive performance and neuronal metabolic resilience**.
- **Prostamax (KEDP) + GH axis + GHK-Cu + TB-500 + BPC-157** when combined aims to **complement soft-tissue repair by targeting organ-specific microenvironment and chromatin-regulatory pathways**.
- **Tα1 provides the "activation layer"** by **boosting dendritic-cell priming, NK-cell cytotoxicity, and interferon-driven Th1 signaling**, while **Thymulin provides the "regulatory layer"** by **tightening thymocyte maturation, stabilizing T-cell activation thresholds, and suppressing excess TNF-α/IL-1β/IL-6**. Together they create a more coordinated, youth-like immune response than either peptide can produce alone.

- **Tadalafil + prostate support:** Daily tadalafil complements **Prostamax and beta-sitosterol** by improving local microcirculation and reducing ischemic contributors to prostate symptoms, reinforcing organ-specific repair strategies.
- **ARA-290 + tadalafil + omega-3**: additive microvascular and endothelial support; monitor blood pressure and symptoms of vasodilation.
- **Spermidine + TRF + NMN + apigenin + resistance training** strengthen autophagy/proteostasis.

Stack Design – Compatibility

Most items occupy **separate physiologic lanes** and are therefore compatible when dosed as listed: cognitive (Semax), mitochondrial supplements (NMN, ALCAR, creatine, Beta-Alanine), and regenerative peptides (TB-500, BPC-157, GHK-Cu, KPV) act in largely distinct domains.

The mitochondrial lane overlap: The mitochondrial / energy supplements are **mechanistically compatible**: **ALCAR** supports mitochondrial substrate entry and acetyl-CoA availability, **CoQ10** enhances ETC efficiency, **creatine** buffers cellular ATP and stabilizes energy supply, and **Beta-Alanine** raises muscle carnosine to improve high-intensity performance; **apigenin** is a polyphenol that modulates signaling pathways (AMPK/SIRT, antioxidant responses); **omega-3s** improve membrane composition, reduce inflammation, and promote mitochondrial resilience. None of these directly block each other's core actions.

The GH lane overlap: Tesamorelin, Ipamorelin, and CJC-1295 modulate GH/IGF-1. These are compatible when coordinated by timing and monitoring. Specifically, using **Tesamorelin in the morning** and **Ipamorelin + CJC-1295 in the evening** is a **compatible strategy**: morning Tesamorelin supports daytime lipolysis and aligns with fasting metabolism, while evening ipamorelin/CJC-1295 augments nocturnal GH pulses and sleep-related recovery, minimizing overlap and preserving a physiologic GH rhythm.

The immune lane overlap: Thymulin and Tα1 are operate at **different, non-overlapping control points of the thymic–immune axis. Thymulin acts upstream as a thymic bioregulatory complex**, while **Tα1 acts downstream as an immune-activation signal**.

The sleep lane overlap: **Epitalon, DSIP , and L-Glycine** operate through **distinct, non-overlapping mechanisms** and they do not share receptor pathways or metabolic bottlenecks.

The annual **Epitalon + Crystagen + Pinealon longevity window** is intentionally run with other peptides paused (except GLP-1s) to create a clear biologic window and reduce confounding interactions.

Stack Design - Conservative Dosing and Cycles

This stack uses **conservative, low weekly dose** of **Retatrutide**, which retain metabolic benefit without promoting additional wight loss. The **Ipamorelin + CJC-1295** and **Tesamorelin** are cycled 12-weeks ON, 6-weeks OFF to **prevent receptor desensitization for GHSR and GHRH pathways** and preserve responsiveness.

By **alternating GLOW (GHK-Cu, TB-500 and BPC-157) and GHK-Cu every 5 weeks** we reduce the uncertainties of continuous TB-500**.**

Overall, several items are **intentionally dosed well below many community-reported maxima to prioritize safety** and tolerability.

Cautions

Informed consent takes on a serious meaning and role in this context.

- **Investigational status and limited evidence:** Many items in this stack are **not FDA approved** and/or are available only as **research chemicals**; long-term safety, dosing, and efficacy data are limited or absent.
- **Off-label and investigational agents: metformin, tadalafil, tesamorelin, and thymosin α1** are being used off-label; **retatrutide** is in late-stage trials but not FDA approved.
- **Community/experimental peptides: Ipamorelin, CJC-1295 no-DAC, GHK-Cu, TB-500, BPC-157, KPV, Semax, Epitalon, Crystagen, Pinealon, Prostamax** have limited human safety data and no established evidence-based dosing.

Specific Warnings

- **Active or recent malignancy, elevated cancer risk: Avoid GH-axis stimulants** (Ipamorelin, CJC-1295, Tesamorelin) and consider avoiding regenerative/angiogenic peptides (GHK-Cu, TB-500, BPC-157) until malignancy risk is evaluated with oncology.
- **Marked baseline IGF-1 elevation or pituitary disease: Do not initiate GH-axis stimulants** until IGF-1 testing and/or endocrine evaluation is complete.

- **Uncontrolled diabetes or insulin deficiency: GLP-1/GIP/glucagon agents and GH stimulants** can alter glycemia; unstable diabetes is a contraindication until glucose control is achieved; type 1 diabetes requires specialist oversight.
- **Significant hepatic impairment:** Metabolic and hepatic-targeting peptides require specialist oversight; **decompensated liver disease is a contraindication**.
- **Severe renal impairment:** Peptide clearance may be altered; **nephrology input** is required for advanced CKD and dose adjustments.
- **Pregnancy and breastfeeding: All peptides and many metabolic agents are contraindicated in pregnancy and lactation**.
- **Active autoimmune disease on immunosuppression: Immune-modulating peptides** may alter immune activity; coordinate with the treating specialist.
- **History of pancreatitis or severe gastroparesis: GLP-1/GIP agents** can exacerbate GI conditions; use only with specialist input.
- **Hypotension and syncope risk: Low-dose daily tadalafil** and other vasodilatory agents can lower systemic vascular resistance; monitor blood pressure, especially with antihypertensives, **alpha-blockers, or nitrates**. Review **nitrate-containing medications, potent CYP3A4 inhibitors (which raise tadalafil levels), and alpha-blockers** before using the stack.
- **Priapism and sensory warnings: PDE5 inhibitors** carry rare risks of **priapism** and sudden visual or auditory changes; seek urgent care if these occur.

Monitoring

Baseline (before starting or continuing): fasting **IGF-1** (use same lab, same assay, same time of day each time), **fasting glucose**, **HbA1c**, **fasting insulin** (HOMA-IR), **CMP** (AST, ALT, bilirubin, creatinine, electrolytes with magnesium), **CBC with differential**, **lipid panel**, **TSH + free T4**, **serum copper + ceruloplasmin + zinc** (for GHK-Cu), **PSA** and age-appropriate cancer screening.

At 3 months, then every 6 months while on therapy: IGF-1, HbA1c, fasting glucose, CMP, lipid panel, CBC; copper, ceruloplasmin, zinc; thyroid panel, PSA. **Off-cycle check:** IGF-1 at 4–6 weeks into washout to confirm return toward baseline.

Action thresholds: IGF-1 >1.5–2× age-adjusted ULN → pause and evaluate; fasting glucose rise >20 mg/dL or HbA1c increase ≥0.3% → reassess metabolic agents; AST/ALT >2× ULN or creatinine rise >30% → pause and investigate.

Possible Adjustments

If IGF-1 exceeds 1.5–2× age-adjusted ULN Ipamorelin+CJC-1295 NODAC and/or Tesamorelin can be reduced.

If fasting glucose and HbA1c rise Retatrutide can be replaced with Tirzepatide.

Biological Mechanisms

Retatrutide — a triple agonist at GLP-1, GIP, and glucagon receptors. By adding **glucagon receptor activation** to incretin signaling, it increases hepatic and systemic **fat oxidation and resting energy expenditure** through enhanced cAMP-mediated lipolysis and mitochondrial substrate switching, while the GLP-1/GIP components maintain **appetite suppression and improved postprandial insulin dynamics**, producing greater net fat loss than incretin agonism alone.

Metformin — a biguanide insulin-sensitizer. It lowers hepatic glucose output primarily by inhibiting **mitochondrial complex I** and activating **AMPK**, which reduces gluconeogenesis and improves peripheral glucose uptake; secondary effects on lipid metabolism, inflammation, and mitochondrial efficiency contribute to its broad metabolic benefits.

Ipamorelin — a selective ghrelin receptor (GHSR) agonist; CJC-1295 NODAC — a GHRH receptor potentiator. Ipamorelin mimics ghrelin to trigger **pulsatile GH release** via GHSR-mediated signaling, while CJC-1295 NODAC prolongs and amplifies GHRH-driven pituitary responsiveness; together they increase nocturnal GH pulse amplitude and frequency, raising downstream **IGF-1** and promoting tissue repair, protein synthesis, and sleep-related recovery when cycled to avoid receptor desensitization.

Tesamorelin — a GHRH analogue. It binds the pituitary GHRH receptor to preferentially increase **endogenous GH secretion**, driving **visceral adipose lipolysis and hepatic fat reduction** through GH-mediated activation of hormone-sensitive lipase and enhanced hepatic lipid oxidation, making it effective for targeted abdominal fat loss in cyclic use.

GHK-Cu — a copper-binding tripeptide (glycyl-histidyl-lysine-copper). It acts as a **matrix-remodeling and pro-repair signal**, upregulating collagen and proteoglycan synthesis, modulating MMP/TIMP balance, and exerting anti-inflammatory and antioxidant effects that accelerate wound healing and improve dermal and soft-tissue regeneration.

TB-500 — a thymosin β4 fragment. It promotes **actin cytoskeleton remodeling, cell migration, and angiogenesis**, facilitating tissue remodeling and repair by enhancing endothelial and progenitor cell mobilization and improving extracellular matrix reorganization across muscle, tendon, and connective tissues.

BPC-157 — a gastric pentadecapeptide. It stimulates **angiogenesis and epithelial restitution**, modulates local growth factor signaling (VEGF, FGF), and reduces pro-inflammatory cytokine activity to support tendon, ligament, and gastrointestinal mucosal healing and to stabilize local microcirculation.

KPV (Lys-Pro-Val) — an anti-inflammatory tripeptide. Derived from alpha-melanocyte-stimulating hormone processing, it **attenuates NF-κB–driven cytokine release**, modulates immune cell activation, and promotes epithelial barrier repair—effects particularly relevant to mucosal (gut) inflammation and systemic inflammatory tone.

Omega-3 (DHA & DPA) — long-chain marine n-3 fatty acids that **integrate into cell and synaptic membranes**, improving **membrane fluidity, receptor function, and neuroprotection**. DHA supplies structural and cognitive support while DPA enhances **specialized pro-resolving mediator (SPM)** production and vascular repair; together they shift eicosanoid signaling toward **anti-inflammatory/pro-resolution pathways**, lower triglycerides, and support endothelial and mitochondrial health.

Thymosin α1 — a thymic peptide immunomodulator. It enhances **T-cell maturation and antigen-presenting cell function**, rebalances innate and adaptive immunity by promoting Th1 responses and regulatory pathways, and improves immune surveillance without broad immunosuppression.

Epitalon — a tetrapeptide acting on pineal and circadian pathways; Crystagen — a thymic peptide. Epitalon modulates **melatonin/circadian signaling** and has been associated in preclinical work with effects on telomere-related pathways and pineal function, while Crystagen supports **thymic peptide signaling** and T-cell homeostasis; together in an annual course they aim to influence circadian biology and aspects of immune aging, though longevity claims remain **speculative**. (Preclinical and limited regional clinical data; not validated by large, randomized trials.)\

Tadalafil — a selective PDE5 inhibitor that raises intracellular cGMP by preventing its breakdown, thereby potentiating nitric-oxide–mediated vasodilation; this **improves endothelial function, microvascular perfusion, and tissue oxygenation**. Chronic PDE5 inhibition also favorably modulates vascular inflammation and arterial stiffness and may influence mitochondrial signaling and perfusion-dependent repair processes in prostate and erectile tissues, making daily low-dose tadalafil a plausible adjunct for healthspan-oriented microcirculatory and metabolic support.

Prostamax (KEDP) — a synthetic tetrapeptide (Lys-Glu-Asp-Pro) from the Khavinson “bioregulator” family, marketed as a **prostate-targeted regulatory peptide** that purportedly **modulates chromatin and gene expression to support prostate tissue**

repair, reduce local inflammation, and improve microcirculation. Clinical evidence is limited and primarily from regional literature, so effects and dosing remain less well-characterized than established peptide therapeutics. (Preclinical and limited regional clinical data; not validated by large, randomized trials.)

Semax — a synthetic peptide derived from ACTH fragment 4–10, delivered intranasally. It **upregulates BDNF and neurotrophic signaling**, modulates monoaminergic neurotransmission, and enhances synaptic plasticity and cognitive resilience, producing short-term improvements in attention, working memory, and stress tolerance when cycled to avoid tolerance.

L Glycine — a nonessential amino acid that **supports collagen synthesis**, serves as an **inhibitory neurotransmitter in the spinal cord and brainstem**, and acts as a key glutathione precursor and **one-carbon metabolism substrate**, thereby contributing to antioxidant defenses and cellular repair. Mechanistically, **oral glycine modulates central NMDA/glycine receptor signaling** and **promotes peripheral vasodilation that lowers core body temperature**, a thermoregulatory change that **facilitates sleep onset**. Clinically, randomized human studies using the commonly studied dose show improved sleep quality, reduced sleep latency, and better next-day alertness.

Beta sitosterol — a plant sterol that **reduces intestinal cholesterol absorption** and may **improve urinary symptoms of benign prostatic hyperplasia**. It works by competing with dietary cholesterol for intestinal uptake, modestly lowering LDL, and by **exerting anti-inflammatory and membrane-modulating effects in prostate tissue**.

β-Nicotinamide Mononucleotide (NMN) — an NAD+ precursor. NMN is converted intracellularly to **NAD+**, restoring redox capacity, supporting **sirtuin-dependent deacetylation**, and improving mitochondrial function and DNA repair pathways that underlie cellular energy metabolism and resilience.

Acetyl-L-Carnitine (ALCAR) — a mitochondrial substrate transporter and acetyl donor. ALCAR facilitates **long-chain fatty-acid transport into mitochondria**, supports acetyl-CoA availability for energy and acetylation reactions, and enhances neuronal and muscular mitochondrial efficiency.

CoQ10 — a mitochondrial electron-transport cofactor, functions as a key electron carrier in Complex I and II of the mitochondrial respiratory chain, improving ATP production efficiency and reducing electron leak and oxidative stress. CoQ10 stabilizes mitochondrial membranes, supports cardiovascular and metabolic function, and provides antioxidant protection.

Creatine HCl — a creatine salt for cellular energy buffering. Creatine replenishes **phosphocreatine stores**, enabling rapid ATP regeneration during high-intensity demands and supporting cellular energetics, osmotic balance, and anabolic signaling.

Beta-Alanine — a carnosine precursor. By increasing intramuscular **carnosine** concentrations, Beta-Alanine **improves intracellular pH buffering during high-intensity exercise**, reducing fatigue and enhancing performance and training capacity; combined with creatine and mitochondrial support, it augments overall energy system resilience.

Apigenin — a naturally occurring flavone that functions as a **potent antioxidant**, **anti-inflammatory agent**, and **mild senolytic signaling modulator**. It influences cellular stress responses by **reducing NF-κB activity**, **activating AMPK and SIRT1 related pathways**, **stabilizing mitochondrial function**, and **improving endothelial and neuronal resilience.** In preclinical models **preserves NAD+ pools** by limiting NAD+ degradation and supporting sirtuin activity

PQQ — a redox-active quinone that stimulates mitochondrial biogenesis. PQQ activates CREB and PGC-1α signaling, increases mitochondrial number, enhances electron-transport efficiency, and reduces oxidative stress. It supports neuronal resilience, improves cellular energy output, and complements NMN/ALCAR/CoQ10 by expanding mitochondrial capacity rather than only improving mitochondrial efficiency.